

A PROJECT SYNOPSIS ON

Watermark-Guided Tamper Localization and Recovery

A Key-Seeded Self-Embedding Fragile Watermarking Scheme
with Joint Localization and Restoration Assessment

SUBMITTED IN PARTIAL FULFILMENT OF THE REQUIREMENTS
FOR THE DEGREE OF BACHELOR OF ENGINEERING
(Computer Engineering / Information Technology)

SUBMITTED BY

Name of Student	Exam Seat No.
[Student 1 — PLACEHOLDER]	[Seat No. — PLACEHOLDER]
[Student 2 — PLACEHOLDER]	[Seat No. — PLACEHOLDER]
[Student 3 — PLACEHOLDER]	[Seat No. — PLACEHOLDER]
[Student 4 — PLACEHOLDER]	[Seat No. — PLACEHOLDER]

UNDER THE GUIDANCE OF

[Guide Name and Designation — PLACEHOLDER]

DEPARTMENT OF COMPUTER ENGINEERING / INFORMATION TECHNOLOGY
AJEENKYA DY PATIL SCHOOL OF ENGINEERING, PUNE
SAVITRIBAI PHULE PUNE UNIVERSITY
ACADEMIC YEAR 2026–27

Project Group ID: [Group ID — PLACEHOLDER]
Domain: Image Processing / Information Security

AJEENKYA DY PATIL SCHOOL OF ENGINEERING, PUNE
DEPARTMENT OF COMPUTER ENGINEERING / INFORMATION
TECHNOLOGY

CERTIFICATE

This is to certify that the project synopsis entitled “**Watermark-Guided Tamper Localization and Recovery: A Key-Seeded Self-Embedding Fragile Watermarking Scheme with Joint Localization and Restoration Assessment**” has been submitted by [Student names — **PLACEHOLDER**] in partial fulfilment of the requirements for the award of the degree of Bachelor of Engineering in Computer Engineering / Information Technology of Savitribai Phule Pune University, during the academic year 2026–27. The work has been carried out under my supervision and guidance, and is approved for submission.

[Guide — **PLACEHOLDER**]

Internal Guide

[Head of Dept. —
PLACEHOLDER]

Dept. of Computer Engg./IT

[Principal —
PLACEHOLDER]

ADYPSOE, Pune

Signature, Student 1

Signature, Student 2

Signature, Student 3

Signature, Student 4

Place: Pune

Date: _____

Abstract

Editing an image convincingly is now a task of seconds, while the formats in which images are stored protect nothing about their content. Passive forensic detectors answer only half of the question a practitioner asks: they may indicate that a region was altered, but they cannot restore the content that was destroyed, and their modern learned instances require annotated training corpora, accelerator hardware at inference time, and produce per-pixel confidence scores that are not explainable under cross-examination. This project develops the active alternative. A deterministic, training-free self-embedding fragile watermarking scheme partitions an image into 8×8 blocks and hides, in the two least-significant bit-planes of every block, a 32-bit keyed authentication tag computed over that block's six most-significant bit-planes, together with a 96-bit compressed recovery descriptor of a spatially distant partner block selected by a key-seeded single-cycle permutation. Verification recomputes each tag; a mismatch localizes tampering at block granularity, and each flagged block is either rebuilt from the descriptor held by its surviving partner or marked explicitly unrecoverable rather than fabricated. On a 32-image corpus, embedding distortion is 43.17 dB PSNR at SSIM 0.9824. Across 768 tamper trials spanning four tamper classes and three tamper ratios, block-level recall is 1.0000 and pixel-level precision 0.9489, and a null condition of 1,802,240 block verifications yields zero false positives. In-region recovery quality remains essentially constant as the tamper ratio grows, from 28.96 dB to 28.38 dB, while whole-image quality falls from 33.79 dB to 17.39 dB, which identifies mapping coverage rather than descriptor fidelity as the dominant constraint on recovery. Every verification decision reduces to a single recomputed keyed hash comparison that any third party holding the key can independently repeat and check bit for bit.

Keywords: fragile watermarking; self-embedding; tamper localization; image authentication; content recovery; block mapping; keyed hashing; image forensics.

Contents

Abstract	2
1 Introduction	4
2 Motivation and Relevance of the Work	4
3 Literature Survey and Market Survey	5
3.1 Passive forensics and learned detectors	5
3.2 Active watermarking: robust, fragile, semi-fragile	5
3.3 Self-embedding and content recovery	6
3.4 The tamper coincidence problem	6
3.5 Reference sharing and fountain-coded allocation	6
3.6 Learned proactive watermarking	7
3.7 Current market survey	8
3.8 Research gap	8
4 Problem Definition	9
5 Objectives	9
6 Scope of the Project	9
7 Proposed System and Methodology	10
7.1 Overview	10
7.2 Embedding	10
7.3 Verification, localization and recovery	11
8 Relevant Mathematics	11
9 List of Modules	12
10 Software and Hardware Requirements	12
11 Expected Outcomes and Results Measured to Date	12
12 Project Plan and Schedule	14
13 Target Publication Venues	15
14 Conclusion	16

1 Introduction

The evidentiary value of a digital image rests on an assumption that is increasingly unsafe: that the pixels presented are the pixels captured. Region-level manipulation — pasting an object from one photograph into another, erasing an object with an inpainting tool, or replacing a cropped region with plausible texture — is available in free software and, with diffusion-based generative editors, requires neither skill nor time. The consequences are concrete rather than hypothetical: a radiological study with a lesion removed, a vehicle-damage photograph submitted with an insurance claim, a surveillance frame entered into a legal record, or a remote-sensing tile used in an environmental dispute all derive their utility from an integrity that the image format itself does not protect.

Two broad families of countermeasures exist. *Passive* (blind) forensics analyses the received image alone, searching for statistical inconsistencies in noise residuals, compression history, colour filter array traces or resampling artefacts; its modern instances are learned [13, 14]. *Active* approaches insert authentication information into the image before distribution, so that verification becomes a decision about self-consistency rather than an inference about provenance; fragile watermarking belongs to this second family [5, 6].

Even a perfect detector answers only half of the practitioner’s question. Knowing precisely *which* pixels of a radiograph were altered does not restore the diagnostic content that was overwritten, and in the passive paradigm restoration is necessarily generative: an inpainting network produces content that is plausible but bears no evidentiary relationship to the original, which is worse than no restoration at all when the artefact is evidence. What is required is a mechanism by which the *original* content — even a coarse approximation of it — survives the attack and can be reinstated. Self-embedding watermarking, in the line of work following Fridrich and Goljan [1], provides exactly this: a compressed representation of each region is hidden elsewhere *within the same image*, so that a spatially localized attack destroys a region but not its backup.

This synopsis describes a complete, reproducible localization-and-recovery pipeline built on that principle which evaluates the two capabilities *together*. Section 2 gives the motivation; Section 3 surveys the literature and the market; Sections 4–6 define the problem, objectives and scope; Section 7 presents the methodology and Section 8 the mathematical model; Sections 9–10 list modules and requirements; Section 11 reports the outcomes measured to date; and Sections 12–14 give the plan of work, target publication venues and conclusion.

2 Motivation and Relevance of the Work

Three observations motivate this work.

First, **detection without restoration is insufficient** in exactly those domains where im-

age integrity matters most. A verifier who learns that a region of a medical scan was altered still cannot say what was there, and a generative reconstruction is not an answer because its content is invented.

Second, **the deployment economics of learned detectors are frequently understated:** annotated corpora whose bias bounds what the model can see, accelerator hardware at inference time that excludes embedded devices and hospital gateways, and outputs that are not explainable under cross-examination. A per-pixel confidence of 0.78 is not an argument. By contrast, every positive detection here is the statement “the 32-bit keyed tag stored in block i does not equal the tag recomputed from block i ’s current content” — verifiable, repeatable and auditable, with no training set, no GPU and no run-to-run variance.

Third, **the literature tends to report localization and recovery separately.** Quoting a healthy in-region reconstruction figure without disclosing what fraction of the flagged region was recoverable at all can flatter a scheme substantially, so this project treats the joint report — localization quality, recoverability rate and recovery fidelity in one table — as a requirement.

The application domains follow: forensic evidence archives, media authentication at ingest, medical imaging integrity in PACS, legal and official document scans, insurance claim verification, and self-authenticating camera or scanner firmware. All share the property that makes an active scheme viable — the acquisition pipeline is under institutional control, so cooperation at capture time is realistic.

3 Literature Survey and Market Survey

3.1 Passive forensics and learned detectors

Passive methods analyse the received image alone. Zhou *et al.* [13] use a two-stream RGB and noise-residual architecture for splice localization, and ManTra-Net [14] uses self-supervised anomaly features to localize forgeries without training on the specific manipulation type. Their structural advantage is decisive and must be acknowledged: they need no cooperation from the acquisition device, which no active scheme can match. Their costs are the annotated corpora, the GPU at inference, the unexplainable output, and the fact that any restoration they offer is hallucinated.

3.2 Active watermarking: robust, fragile, semi-fragile

Early watermarking optimised *robustness*: Cox *et al.* [2] embed the mark in perceptually significant transform coefficients so that ownership survives compression and filtering. Authentication inverts the requirement — the mark must be fragile, because its failure is the signal. Wong and Memon [5] developed block-wise fragile schemes with per-block signatures in the least-significant bits, giving localized, blind verification. The weakness of pure

fragility is that it cannot distinguish a malicious splice from a benign re-save; Lin and Chang [4] answered this with an invariant relation between DCT coefficient pairs that quantization preserves and manipulation breaks, establishing the semi-fragile pattern. The present work is deliberately fully fragile, and treats the semi-fragile extension as future scope.

3.3 Self-embedding and content recovery

The step from detection to recovery is due to Fridrich and Goljan [1]: embed a compressed version of each block inside the image itself. Their triad — block-wise compression, key-dependent mapping to a distant carrier, LSB embedding — is the direct ancestor of this design. Lin *et al.* [8] added a hierarchical multi-level block structure verifying at several granularities, improving localization resolution and the robustness of the tamper decision.

Security analysis runs alongside. Holliman and Memon [3] showed that if a block’s watermark depends only on that block’s own content, an adversary can collage authentic blocks into a counterfeit in which every block verifies. Fridrich [7] analysed localizing fragile marks in the same spirit. The practical consequence, adopted here without modification, is that the authentication tag must bind block content, block index and an image-level identifier under a secret key.

3.4 The tamper coincidence problem

A one-to-one backup allocation has a characteristic failure: a block and the block holding its recovery data may be destroyed by the same attack. Aminuddin and Ernawan named this the *tamper coincidence problem* and, in AuSR3 [16], addressed it by mapping each block’s recovery data to the most distant available location; the earlier AuSR1 [15] established the family’s 2×2 recovery granularity. **The distant-mapping principle is theirs, not ours**, and this synopsis does not claim it. What this project contributes is the specific construction: a key-seeded permutation built directly in cycle notation, so that bijectivity and the absence of short cycles hold unconditionally rather than being checked after the fact, combined with an enforced minimum-separation repair pass.

3.5 Reference sharing and fountain-coded allocation

Zhang *et al.* [9] reframed payload allocation so that reference information is *shared* across blocks and reconstruction becomes the solution of a linear system, lifting the tolerable tamper ratio above the naive one-to-one limit. Korus and Dziech [10] added content-adaptive self-embedding with a formal rate–distortion analysis. This matters directly to the present design and is stated plainly in Section 11: a one-to-one mapping fails all-or-nothing per block, whereas shared allocations degrade gracefully, which is why Korus and Dziech report far higher whole-image reconstruction at high damage levels than this scheme achieves.

3.6 Learned proactive watermarking

A recent family embeds a learned watermark so that a trained network both localizes and recovers; EditGuard [17] is representative, jointly optimising an invertible watermarking network for localization and copyright protection. “Active watermarking detects but does not recover” is therefore no longer an accurate description of the field, and this project does not claim better accuracy than these systems. The distinction offered is auditability: a decision here is one recomputed keyed hash, reproducible bit for bit by any third party holding the key, whereas a learned localizer’s decision is a trained network’s output and cannot be recomputed without its weights.

Table 1: Summary of the surveyed literature.

Ref.	Author(s), Year	Technique	Limitation / gap
[2]	Cox <i>et al.</i> , 1997	Spread-spectrum robust watermark in significant coefficients	Robustness is the inverse of authentication; cannot signal change
[1]	Fridrich & Goljan, 1999	Origin of self-embedding; compressed backup at a distant carrier	Recovery approximate; no keyed binding of block index
[3]	Holliman & Memon, 2000	Counterfeiting attack on block-wise independent marks	Attack, not scheme; forces index and image-ID binding
[4]	Lin & Chang, 2000	Semi-fragile authentication via DCT coefficient-pair invariants	Tolerant to compression, so weaker localization precision
[5]	Wong & Memon, 2001	Block-wise fragile signatures in LSBs; blind, localized	Authenticates only; no content recovery
[7]	Fridrich, 2002	Security analysis of localizing fragile marks	Analysis; no recovery mechanism
[8]	Lin <i>et al.</i> , 2005	Hierarchical multi-level verification	Higher payload cost per block
[9]	Zhang <i>et al.</i> , 2011	Reference sharing; reconstruction as a linear system	Higher decode complexity; needs global solve
[10]	Korus & Dziech, 2013	Content-adaptive self-embedding, rate-distortion analysis	Substantially more complex allocation
[15]	Aminuddin & Ernawan, 2022	AuSR1: 2×2 recovery, LSB shifting	LSB shifting perturbs bits an MSB-authenticating tag must cover
[16]	Aminuddin & Ernawan, 2023	AuSR3: distant block mapping against tamper coincidence	Comparable per-class recovery figures not citable without re-implementation
[17]	Zhang <i>et al.</i> , 2024	Learned proactive watermark; localizes and recovers	Requires training and model weights; decision not independently recomputable

3.7 Current market survey

Deployed products in this space fall into three groups, none of which offers localization and recovery together from within the image itself.

Provenance metadata. The C2PA Content Credentials standard, adopted by several camera and editing-software vendors, attaches a cryptographically signed manifest recording an asset’s origin and edit history. Its strength is standardization and industry backing. Its weakness for this problem is structural: the manifest lives *beside* the pixels, so re-encoding, a screenshot, or a metadata strip removes it entirely, and it can never restore content — it attests to history rather than protecting it.

Controlled-capture services. Commercial offerings such as Truepic authenticate images at the moment of capture through a proprietary pipeline. They give strong provenance where they are used, but require the vendor’s software and infrastructure, and again do not recover destroyed content.

Forensic analysis tools. Analyst-facing products such as Amped Authenticate, and free services in the FotoForensics mould, apply passive heuristics — error-level analysis, noise and CFA inconsistency, metadata review — to images with no prior cooperation. Their advantage is that they work on arbitrary images. Their limitations are that results require expert interpretation, carry no ground truth, and again offer no restoration.

Whole-file digital signatures, as standardized for DICOM medical imaging, detect that a file changed but neither localize the change to a region nor recover it.

The gap in the market mirrors the gap in the literature: existing products either attest to provenance externally or analyse pixels after the fact, and none embeds recoverable content inside the image so that a damaged region can be restored from what survived.

3.8 Research gap

The gap addressed here is deliberately narrow, and stating it narrowly is part of the contribution. It is *not* the distant-mapping idea, which is credited to AuSR3 [16]. It is *not* superiority over learned proactive watermarking [17], which already localizes and recovers and against which no accuracy claim is made. The gap is threefold. First, **determinism and auditability**: a construction in which every positive detection reduces to one recomputed keyed hash comparison that a third party holding the key can repeat bit for bit, with no training data, no GPU and no run-to-run variance. Second, **an explicit keyed construction** with each field binding stated and analysed, including what each binding defeats. Third, **joint evaluation**: imperceptibility, localization and recovery reported together, with the recoverability rate and the unrecoverable fraction disclosed rather than excluded, across four tamper classes and three tamper ratios against exact ground-truth masks.

4 Problem Definition

Given a digital image stored in a lossless container and a secret key held by the authenticating party, embed authentication and recovery information into the image itself such that: any subsequent region-level modification is localized to the block grid without reference to the original image or any external database; the original content of the modified region is approximately restored from information that survived within the same image; and any region that cannot be restored is identified as such rather than reconstructed by inference. The embedding must be perceptually transparent, the verification decision must be deterministic and independently recomputable, and the entire pipeline must run on a commodity CPU without trained parameters.

5 Objectives

- 01.** Design a per-block payload that fits entirely within two least-significant bit-planes, so that embedding distortion is analytically bounded, and achieve mean imperceptibility of at least 40 dB PSNR over a standard corpus.
- 02.** Construct a key-seeded block mapping that is provably bijective, free of fixed points and short cycles, and enforces a minimum spatial separation between a block and its recovery partner.
- 03.** Localize region-level tampering at block granularity with block-level recall of 1.0 across four distinct tamper classes and three tamper ratios.
- 04.** Produce zero false positives on unmodified watermarked images, verified over a null condition of more than 10^6 independent block verifications.
- 05.** Reconstruct flagged regions from partner-held descriptors, report an explicit recoverability rate, and mark unrecoverable regions rather than fabricating content.
- 06.** Deliver a reproducible evaluation grid and working software, gated by automated checks that must pass before any result is published.

6 Scope of the Project

In scope. Still images, greyscale and three-channel colour, at 8 bits per channel, stored in lossless containers (PNG, BMP, TIFF). Block sizes 8 and 4. Two evaluated recovery descriptor designs. Four tamper classes — copy-paste splicing, object removal by inpainting, crop-and-refill, and random block corruption — at three tamper ratios. A 32-image evaluation corpus. A local web application and a demonstration interface over the same pipeline.

Out of scope. Video and audio. Robustness to lossy re-encoding, resizing, rotation or filtering — the scheme is fully fragile by design. Adversaries in possession of the secret key, against whom no active scheme can defend. Real-time embedding in camera firmware,

which is an engineering integration rather than a research question. Comparison against learned detectors on their own accuracy terms.

7 Proposed System and Methodology

7.1 Overview

The system has four decoupled stages: embedding at or near acquisition; tampering, which is simulated by the evaluation harness; verification and localization at any later point; and recovery. Verification requires only the received image, the key and the image identifier — no original, no reference database, no network call and no model weights.

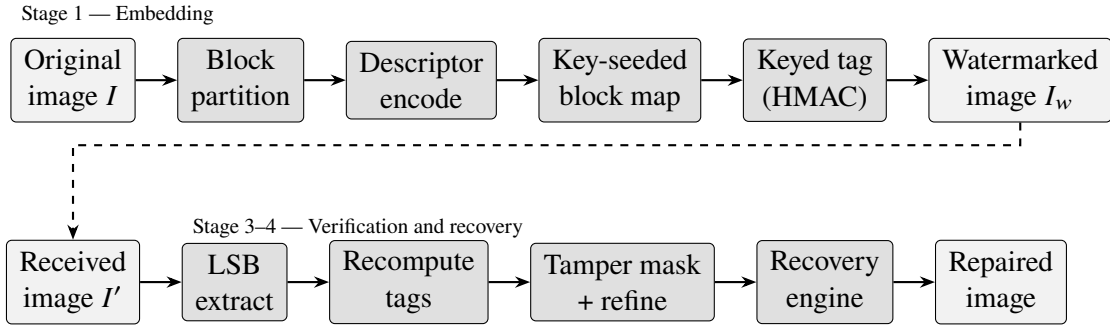


Figure 1: System architecture. The key K_s and image identifier seed both the block map and the tag; the dashed path carries the distributed image, which may be tampered before it reaches verification.

7.2 Embedding

The image is partitioned into non-overlapping $B \times B$ blocks in raster order, with $B = 8$ by default, giving $K = (M/B)(N/B)$ blocks. Dimensions that are not a multiple of B are cropped rather than padded, so that no synthetic content is ever authenticated. The two least-significant bit-planes of each block provide $2B^2 = 128$ bits of capacity, allocated as a 32-bit authentication tag and a 96-bit recovery descriptor.

All hashed and compressed content is computed from the block’s six most-significant bit-planes, obtained by the projection $\text{msb}(a) = a \wedge 0\text{xFC}$. This is the property that makes the scheme single-pass: because embedding writes only to the two LSBs, it cannot alter what was signed, so no iterative embed-and-verify loop is required.

The recovery descriptor compresses a block to 96 bits. The primary variant applies a level-shifted orthonormal two-dimensional DCT and retains the first twelve zig-zag coefficients, each quantized to a signed 8-bit value using per-band step sizes taken from the JPEG quality-50 luminance table, giving $12 \times 8 = 96$ bits exactly. A second, low-complexity variant stores a 4×4 array of 2×2 block means at 6 bits each, requiring no transform.

Descriptors are then permuted: block i ’s descriptor is stored in a distant partner block, and the tag written into block i binds the descriptor that block i physically carries. The order

matters — the descriptor is placed first, and the tag that authenticates it is computed second.

7.3 Verification, localization and recovery

Verification extracts the 128 stored bits from each block, splits them into the stored tag and the stored descriptor, recomputes the tag over the block’s current most-significant content *and the descriptor bits actually present*, and flags a mismatch. Per-channel decisions are combined by logical OR, so a tamper touching any single channel flags the block. A neighbourhood refinement pass then fills isolated negatives, using a threshold scaled by the number of valid neighbours so that blocks at the image border are not penalised.

Recovery evaluates three sets on a frozen copy of the mask, before any pixel is written: the flagged set $\mathcal{T}$, the subset $\mathcal{R}$ whose partner is intact and which can therefore be rebuilt, and the subset $\mathcal{U}$ whose partner is also flagged. Blocks in $\mathcal{R}$ are decoded from the surviving descriptor. Blocks in $\mathcal{U}$ are marked with a flat value — never inpainted, never silently left in place — so that a viewer can always tell reconstructed content from content that could not be recovered.

8 Relevant Mathematics

Let $I \in \{0, \dots, 255\}^{M \times N}$ be the host image, K_s a secret key, and ID an image identifier. Write b_i for the i -th block, $i = 0, \dots, K - 1$, and $c(i)$ for its centre coordinate.

Bit budget. For block size B ,

$$\text{cap}(B) = 2B^2, \quad t(B) = \min(32, \text{cap}(B)/2), \quad d(B) = \text{cap}(B) - t(B), \quad (1)$$

so $B = 8$ gives (128, 32, 96) bits of capacity, tag and descriptor respectively.

Most-significant projection.

$$\text{MSB}(a) = a \wedge 0\text{xFC}, \quad (2)$$

which zeroes the two least-significant bits and is idempotent.

Block map. The mapping m is a permutation of $\{0, \dots, K - 1\}$ satisfying

$$(R1) \ m \text{ bijective}, \quad (R2) \ \|c(i) - c(m(i))\|_2 \geq d_{\min}, \quad (R3) \ m \text{ is a single } K\text{-cycle}, \quad (3)$$

with $d_{\min} = \frac{1}{4} \min(M, N)$. R3 is obtained by construction rather than by rejection sampling: a cryptographically seeded Fisher–Yates shuffle produces a cycle order, and m is read off it directly, which makes fixed points and mutual pairs impossible.

Descriptor. With C_i the DCT of the level-shifted projected block and $\sigma(\ell)$ the zig-zag

index,

$$\hat{q}_{i,\ell} = \text{clip}\left(\text{round}\left(\frac{C_i(\sigma(\ell))}{\Delta_\ell}\right), -128, 127\right), \ell = 1, \dots, 12. \quad (4)$$

No retained coefficient can clip: by Cauchy–Schwarz $|C| \leq 1024$ at $B = 8$, and every step size is large enough that the quantized magnitude stays inside the signed 8-bit range.

Authentication tag.

$$a_i = \text{Trunc}_{32}\left(\text{HMAC}_{K_s}(\text{ID} \parallel i \parallel \text{MSB}(b_i) \parallel r_{m^{-1}(i)})\right), \quad (5)$$

using SHA-256 [12]. Each bound field defeats a specific attack: $\text{MSB}(b_i)$ makes the tag independent of the bits embedding will write; i and ID defeat the collage attack of [3]; and $r_{m^{-1}(i)}$, the descriptor this block carries for another, prevents an attacker from replacing recovery data without detection.

Recoverability rate. With $\mathcal{T}$ the flagged set and $\mathcal{U} = \{i \in \mathcal{T} : m(i) \in \mathcal{T}\}$,

$$\rho = 1 - \frac{|\mathcal{U}|}{|\mathcal{T}|}. \quad (6)$$

Complexity. Embedding and verification each require one keyed hash and one $B \times B$ transform per block, plus $O(K)$ for the mapping, giving $O(MN)$ time overall with no iteration, no optimization and no training. The problem is therefore of class P; no NP-hard sub-problem arises anywhere in the pipeline.

9 List of Modules

10 Software and Hardware Requirements

11 Expected Outcomes and Results Measured to Date

The evaluation grid comprises **1,184 rows**: 768 main-grid tamper trials (32 images $\times$ 4 tamper classes $\times$ 3 tamper ratios $\times$ 2 descriptor variants), 320 null-condition rows (32 images $\times$ 2 variants $\times$ 5 keys), and 96 block-size ablation rows. The corpus is 32 colour images — 8 from USC-SIPI [18] at 512×512 and 24 from the Kodak set — with every file pinned by SHA-256 so the corpus reproduces byte for byte. An automated gate reports 23 of 23 checks passing before any figure below is quoted.

Three results deserve comment.

Localization. Block-level recall is exactly 1.0000 across all 768 main-grid trials. Block-level *precision* is deliberately not headlined: under the ground-truth rule used here — a block is tampered if the tamper region covers any of its pixels — a block-level false positive is structurally impossible, so that metric would read 1.000 regardless of detector quality. The informative figure is pixel-level precision, 0.9489, whose shortfall from 1.0 is entirely

Table 2: Functional modules and their responsibilities.

Module	Responsibility
Payload foundation	Key derivation and domain-separated subkeys, bit budget, block geometry, MSB projection, keyed tag computation, descriptor encoding and decoding, LSB packing. Imported by every other module.
Block mapper	Builds the key-seeded single-cycle permutation, enforces the minimum-separation constraint, and verifies R1–R3 before returning.
Embedder	Stage 1. Computes descriptors, permutes them, computes the binding tags, writes both into the two LSB planes; asserts that the MSB planes are unchanged.
Detector and localizer	Stage 3. Extracts stored payload, recomputes tags, produces the block-level tamper mask, applies neighbourhood refinement, expands to a pixel mask, and exposes a per-block audit record.
Recovery engine	Stage 4. Partitions the flagged set into recoverable and unrecoverable, rebuilds the former from partner descriptors, marks the latter, and reports ρ .
Tamper harness	Generates the four tamper classes with exact ground-truth masks and the achieved tamper ratio.
Metrics and evaluation	PSNR, SSIM, confusion counts, localization scores, in-region recovery metrics, and the CSV result grid.
Automated gate	Validates the result grid against expected bands and refuses to emit tables or figures if any check fails.
Applications	A local web application and a demonstration interface over the same pipeline, with no duplicated numerics.

Table 3: Requirements.

Category	Requirement
Language	Python 3.10 or newer. Cryptographic primitives from the standard library (hashlib, hmac, struct); no third-party cryptography.
Core libraries	NumPy 2.1.2, OpenCV (headless) 5.0.0.93, scikit-image 0.26.0, Pillow 10.4.0 — all pinned exactly for reproducibility.
Interface libraries	Flask for the local web application; Streamlit 1.61.1 for the demonstration interface; Matplotlib 3.11.1 for figures.
Storage	SQLite for the local protected-image library.
Document build	tectonic and pandoc (optional; required only to rebuild the paper and this synopsis).
Machine learning	None. No framework, no trained parameters, no model artefact to version.
Processor	Any commodity x86-64 or ARM CPU. Execution is single-threaded.
Accelerator	No GPU required at any stage — for embedding, verification or recovery.
Performance	A 512×512 image is processed end-to-end in tens of milliseconds on a commodity CPU.
Operating system	Platform independent (developed and tested on Windows; no OS-specific dependency).

Table 4: Imperceptibility of watermarked images (32-image corpus mean).

Descriptor variant	PSNR (dB)	SSIM
Variant A (DCT, 12 coefficients)	43.17	0.9824
Variant B (mean-pooled)	44.23	0.9844

Table 5: Localization and recovery against tamper ratio α (Variant A, averaged over the four tamper classes).

Quantity	$\alpha = 10\%$	$\alpha = 25\%$	$\alpha = 50\%$
Block-level recall	1.0000	1.0000	1.0000
Pixel-level precision (grid mean)		0.9489	
Recoverability rate ρ	0.9590	0.8036	0.5335
In-region recovery PSNR (dB)	28.96	28.44	28.38
Whole-image PSNR, unmarked (dB)	33.79	23.87	17.39

the block-quantization penalty: a flagged 8×8 block is reported whole even when only part of it was altered.

The null condition. Zero false-positive blocks were observed across 1,802,240 independent block verifications, giving a rule-of-three 95% upper bound of 1.66×10^{-6} on the per-block false-positive rate. This is reported as a determinism and implementation-consistency check across images, variants and keys — not as a statistical bound on the 2^{-32} cryptographic false-accept probability, which this experiment does not measure at all.

Coverage dominates recovery quality. In-region recovery PSNR is essentially flat as the tamper ratio grows (28.96 to 28.38 dB) while whole-image PSNR collapses (33.79 to 17.39 dB). The two answer different questions: the first asks how good a reconstructed block is, which the descriptor determines independently of how many blocks needed reconstruction; the second asks how good the picture is, which depends on how much of it had to be reconstructed. This is direct measured evidence that mapping coverage, not descriptor fidelity, is the dominant lever — and it identifies reference sharing as the highest-value upgrade. We state plainly that a one-to-one mapping falls well short of reference-sharing and fountain-coded schemes at high tamper ratios [9, 10].

A block-size ablation quantifies the resolution/payload trade-off directly: reducing B from 8 to 4 raises pixel-level precision from 0.9397 to 0.9736, at a cost of 2.9 dB of in-region recovery fidelity (29.66 to 26.80 dB), because the per-block payload falls from 128 to 32 bits and the authentication tag halves.

12 Project Plan and Schedule

Probable date of completion: [Date, as per departmental academic calendar — PLACEHOLDER].

Table 6: Plan of work. Phases 1–6 are complete; the remainder is scheduled.

Ph.	Activity	Deliverable	Status
1	Literature survey and problem formulation	Survey, gap statement	Complete
2	Scheme design: bit budget, tag construction, block map	Design specification	Complete
3	Implementation of the core pipeline	Embed / detect / recover modules	Complete
4	Tamper harness and metrics	Four tamper classes, exact masks	Complete
5	Full experimental grid and validation	1,184-row result grid, 23/23 gate	Complete
6	Applications and adversarial testing	Web application; two defects found and fixed	Complete
7	Paper preparation and internal review	IEEE-format paper (Annexure A)	In progress
8	Variant C consolidation into the reported grid	Updated tables and figures	Planned
9	Reference-sharing extension (Future Scope 1)	Prototype and comparison	Planned
10	Final report, demonstration and submission	Report, presentation, viva	Planned

13 Target Publication Venues

The work is intended for submission to a peer-reviewed venue in image forensics or multimedia security. Two candidate venues, both appropriate in scope:

1. **IEEE International Workshop on Information Forensics and Security (WIFS)** — the natural venue for authentication and forensic watermarking, and the community that publishes the tamper-localization literature surveyed above.
2. **IEEE International Conference on Image Processing (ICIP)**, which has a broad image-processing scope and a long-established watermarking and forensics track; the Fridrich and Goljan self-correcting-images paper that founds this line of work appeared there.

Two further options, should a national or journal venue be preferred: the *National Conference on Communications* (NCC), and the *EURASIP Journal on Information Security*. Submission deadlines vary by edition and are to be confirmed against the current call for papers before selection.

A complete IEEE-format manuscript has already been prepared and is attached as Annexure A, so submission is a matter of venue selection and formatting rather than of new writing.

14 Conclusion

This project delivers a deterministic, training-free self-embedding fragile watermarking scheme that localizes tampering at block granularity and restores a low-frequency approximation of the destroyed content from information carried within the image itself. Measured over a 1,184-row experimental grid, it embeds at 43.17 dB PSNR and SSIM 0.9824, achieves block-level recall of 1.0000 and pixel-level precision of 0.9489, and returns zero false positives across 1,802,240 block verifications. Its recovery coverage degrades toward $1 - \alpha$ for dispersed attacks — the structural limit of one-to-one backup allocation — and this synopsis states that limitation rather than concealing it; adopting a reference-sharing allocation is the identified next step. The property that distinguishes the design is not accuracy but accountability: every verification decision is a single recomputed keyed hash comparison, reproducible bit for bit by any third party holding the key, which is precisely what an evidentiary application requires and what a learned detector cannot offer.

References

- [1] J. Fridrich and M. Goljan, “Protection of digital images using self-embedding,” in *Proc. Symp. Content Security and Data Hiding in Digital Media*, Newark, NJ, USA, May 1999.
- [2] I. J. Cox, J. Kilian, F. T. Leighton, and T. Shamoon, “Secure spread spectrum watermarking for multimedia,” *IEEE Trans. Image Process.*, vol. 6, no. 12, pp. 1673–1687, Dec. 1997.
- [3] M. Holliman and N. Memon, “Counterfeiting attacks on oblivious block-wise independent invisible watermarking schemes,” *IEEE Trans. Image Process.*, vol. 9, no. 3, pp. 432–441, Mar. 2000.
- [4] C.-Y. Lin and S.-F. Chang, “Semi-fragile watermarking for authenticating JPEG visual content,” in *Proc. SPIE 3971, Security and Watermarking of Multimedia Contents II*, 2000, pp. 140–151.
- [5] P. W. Wong and N. Memon, “Secret and public key image watermarking schemes for image authentication and ownership verification,” *IEEE Trans. Image Process.*, vol. 10, no. 10, pp. 1593–1601, Oct. 2001.
- [6] C. Rey and J.-L. Dugelay, “A survey of watermarking algorithms for image authentication,” *EURASIP J. Appl. Signal Process.*, vol. 2002, no. 6, pp. 613–621, 2002.
- [7] J. Fridrich, “Security of fragile authentication watermarks with localization,” in *Proc. SPIE 4675, Security and Watermarking of Multimedia Contents IV*, San Jose, CA, USA, Jan. 2002, pp. 691–700.

- [8] P.-L. Lin, C.-K. Hsieh, and P.-W. Huang, “A hierarchical digital watermarking method for image tamper detection and recovery,” *Pattern Recognit.*, vol. 38, no. 12, pp. 2519–2529, Dec. 2005.
- [9] X. Zhang, S. Wang, Z. Qian, and G. Feng, “Reference sharing mechanism for watermark self-embedding,” *IEEE Trans. Image Process.*, vol. 20, no. 2, pp. 485–495, Feb. 2011.
- [10] P. Korus and A. Dziech, “Efficient method for content reconstruction with self-embedding,” *IEEE Trans. Image Process.*, vol. 22, no. 3, pp. 1134–1147, Mar. 2013.
- [11] Z. Wang, A. C. Bovik, H. R. Sheikh, and E. P. Simoncelli, “Image quality assessment: From error visibility to structural similarity,” *IEEE Trans. Image Process.*, vol. 13, no. 4, pp. 600–612, Apr. 2004.
- [12] National Institute of Standards and Technology, “Secure Hash Standard (SHS),” FIPS Publication 180-4, Aug. 2015.
- [13] P. Zhou, X. Han, V. I. Morariu, and L. S. Davis, “Learning rich features for image manipulation detection,” in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, Salt Lake City, UT, USA, Jun. 2018.
- [14] Y. Wu, W. AbdAlmageed, and P. Natarajan, “ManTra-Net: Manipulation tracing network for detection and localization of image forgeries with anomalous features,” in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, Long Beach, CA, USA, Jun. 2019.
- [15] A. Aminuddin and F. Ernawan, “AuSR1: Authentication and self-recovery using a new image inpainting technique with LSB shifting in fragile image watermarking,” *J. King Saud Univ. – Comput. Inf. Sci.*, 2022.
- [16] A. Aminuddin and F. Ernawan, “AuSR3: A new block mapping technique for image authentication and self-recovery to avoid the tamper coincidence problem,” *J. King Saud Univ. – Comput. Inf. Sci.*, vol. 35, no. 9, art. 101755, 2023.
- [17] X. Zhang, R. Li, J. Yu, Y. Xu, W. Li, and J. Zhang, “EditGuard: Versatile image watermarking for tamper localization and copyright protection,” in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, 2024.
- [18] University of Southern California, Signal and Image Processing Institute, “The USC-SIPI Image Database.” [Online]. Available: <https://sipi.usc.edu/database/>

Annexure A. Full IEEE-format manuscript, prepared in the IEEEtran conference format and compiled to IEEE_Paper .pdf (15 pages), attached separately.